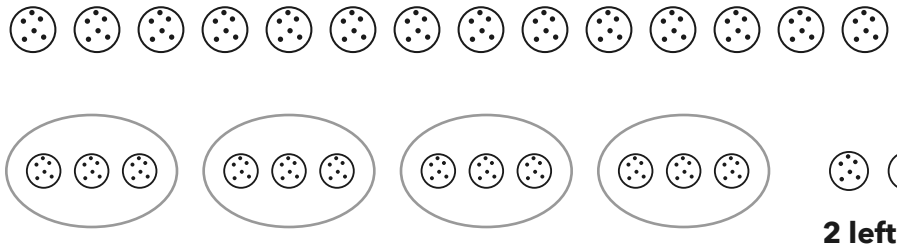
Division with Remainders

Share fairly. What is left over is the remainder.

14 cookies on 4 plates



2 left over

14 ÷ 4 = 3 R 2 3 on each plate, remainder 2. Check: 4 × 3 + 2 = 14.

Use a times fact: the biggest one that fits

23 ÷ 5:

5 × 4 = 20 (5 × 5 = 25 is too big)

23 − 20 = 3

23 ÷ 5 = 4 R 3

What does the remainder mean?

14 kids, 4 in each car: 3 cars are full and 2 kids still need a ride, so 4 cars.

14 cookies shared by 4 friends: each gets 3, and 2 are left.

Your turn!

$3 \times \boxed{5} = \boxed{15}$

$17 \div 3 = \boxed{5}\ \text{R}\ \boxed{2}$

For teachers

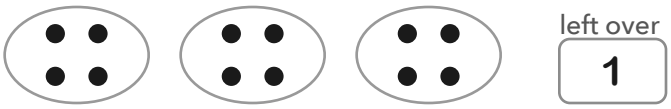
Goal: divide with a remainder and decide what it means in a story (4.OA.3). Share real objects first. Say "What is the biggest times fact that fits? What is left?" The remainder is always less than the divisor.

Division with Remainders: Practice

Share fairly. What is left over?

Part 1 Share on plates Share the dots on the plates. Write what is left over.

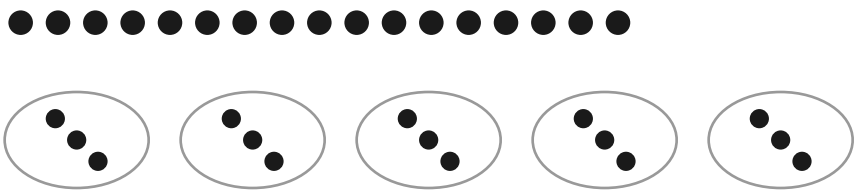
1)



left over: 1

13 ÷ 3 = 4 R 1

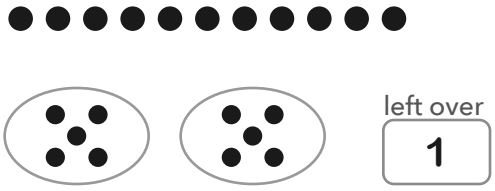
2)



left over: 2

17 ÷ 5 = 3 R 2

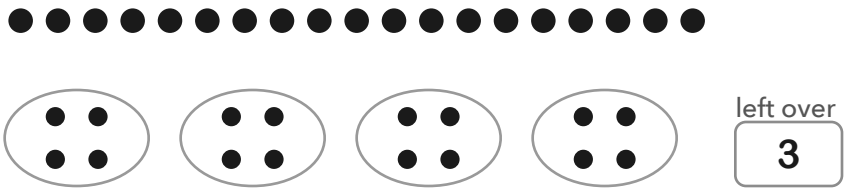
3)



left over: 1

11 ÷ 2 = 5 R 1

4)



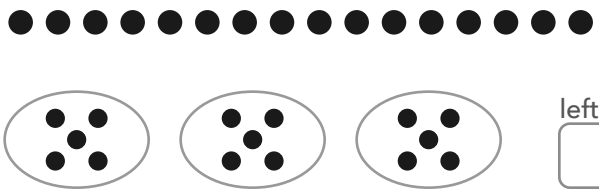
left over: 3

19 ÷ 4 = 4 R 3

Division with Remainders: Practice (continued)

Part 1 Share on plates (continued)

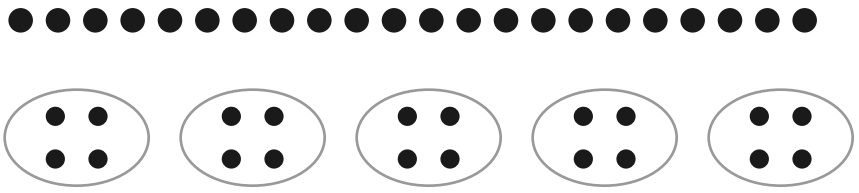
5)



left over: 1

16 ÷ 3 = 5 R 1

6)



left over: 2

22 ÷ 5 = 4 R 2

Part 2 Make groups Circle the groups. Count the leftovers.

7) **Circle groups of 4.**



3 groups of 4, 2 left over

8) **Circle groups of 3.**



5 groups of 3, 2 left over

Division with Remainders: Practice (continued)

Part 2 Make groups (continued)

9) **Circle groups of 5.**



4 groups of 5, 3 left over

10) **Circle groups of 2.**



6 groups of 2, 1 left over

11) **Circle groups of 6.**



3 groups of 6, 2 left over

12) **Circle groups of 4.**



4 groups of 4, 3 left over

Part 3 Use a times fact Find the biggest fact that fits, then what is left.

13) **29 ÷ 4: the biggest fact up to 29**

4 × 7 = 28

29 − 28 = 1

so 29 ÷ 4 = 7 R 1

14) **38 ÷ 5: the biggest fact up to 38**

5 × 7 = 35

38 − 35 = 3

so 38 ÷ 5 = 7 R 3

Division with Remainders: Practice (continued)

Part 3 Use a times fact (continued)

15) 45 ÷ 7: the biggest fact up to 45

7 × 6 = 42

45 − 42 = 3

so 45 ÷ 7 = 6 R 3

16) 26 ÷ 3: the biggest fact up to 26

3 × 8 = 24

26 − 24 = 2

so 26 ÷ 3 = 8 R 2

17) 53 ÷ 6: the biggest fact up to 53

6 × 8 = 48

53 − 48 = 5

so 53 ÷ 6 = 8 R 5

18) 61 ÷ 8: the biggest fact up to 61

8 × 7 = 56

61 − 56 = 5

so 61 ÷ 8 = 7 R 5

Part 4 Divide No help this time.

19) 17 ÷ 5 = 3 R 2

20) 23 ÷ 4 = 5 R 3

21) 31 ÷ 6 = 5 R 1

22) 44 ÷ 7 = 6 R 2

23) 19 ÷ 2 = 9 R 1

24) 38 ÷ 9 = 4 R 2

Division with Remainders: Practice (continued)

Part 4 Divide (continued)

25) $27 \div 8 = 3 \text{ R } 3$

26) $50 \div 6 = 8 \text{ R } 2$

27) $35 \div 4 = 8 \text{ R } 3$

28) $47 \div 5 = 9 \text{ R } 2$

29) $62 \div 7 = 8 \text{ R } 6$

30) $29 \div 3 = 9 \text{ R } 2$

Part 5 Bigger numbers No help this time.

31) $73 \div 8 = 9 \text{ R } 1$

32) $58 \div 9 = 6 \text{ R } 4$

33) $66 \div 7 = 9 \text{ R } 3$

34) $85 \div 9 = 9 \text{ R } 4$

35) $79 \div 8 = 9 \text{ R } 7$

36) $52 \div 6 = 8 \text{ R } 4$

Division with Remainders: Practice (continued)

Part 6 What is left?

37) $39 \div 5 = 7 \text{ R } 4$

38) $15 \div 7 = 2 \text{ R } 1$

39) $22 \div 4 = 5 \text{ R } 2$

40) $11 \div 7 = 1 \text{ R } 4$

41) $70 \div 9 = 7 \text{ R } 7$

42) $25 \div 6 = 4 \text{ R } 1$

43) $36 \div 7 = 5 \text{ R } 1$

44) $53 \div 8 = 6 \text{ R } 5$

45) $17 \div 6 = 2 \text{ R } 5$

46) $87 \div 9 = 9 \text{ R } 6$

47) $23 \div 6 = 3 \text{ R } 5$

48) $38 \div 6 = 6 \text{ R } 2$

Division with Remainders: Practice (continued)

Part 6 What is left? (continued)

49) $59 \div 8 = 7 \text{ R } 3$

50) $32 \div 6 = 5 \text{ R } 2$

51) $41 \div 9 = 4 \text{ R } 5$

52) $29 \div 6 = 4 \text{ R } 5$

53) $65 \div 8 = 8 \text{ R } 1$

54) $43 \div 5 = 8 \text{ R } 3$

55) $38 \div 7 = 5 \text{ R } 3$

56) $29 \div 4 = 7 \text{ R } 1$

57) $43 \div 7 = 6 \text{ R } 1$

58) $27 \div 4 = 6 \text{ R } 3$

59) $18 \div 5 = 3 \text{ R } 3$

60) $59 \div 6 = 9 \text{ R } 5$

Division with Remainders: Practice (continued)

Part 6 What is left? (continued)

61) $27 \div 6 = 4 \text{ R } 3$

62) $32 \div 9 = 3 \text{ R } 5$

63) $43 \div 8 = 5 \text{ R } 3$

64) $33 \div 8 = 4 \text{ R } 1$

65) $14 \div 4 = 3 \text{ R } 2$

66) $42 \div 9 = 4 \text{ R } 6$

67) $66 \div 8 = 8 \text{ R } 2$

68) $33 \div 6 = 5 \text{ R } 3$

69) $18 \div 8 = 2 \text{ R } 2$

70) $60 \div 7 = 8 \text{ R } 4$

71) $41 \div 6 = 6 \text{ R } 5$

72) $60 \div 9 = 6 \text{ R } 6$

Division with Remainders: Practice (continued)

Part 7 Remember? Problems from earlier packets.

73) $43 \times 6 =$ **258**

74) 7,281 to the nearest thousand = **7,000**

75) $6{,}304 - 2{,}857 =$ **3,447**

76) in 3,519 the 5 is worth **500**

77) each picture = 10: 9 pictures = **90**

78) 7 m by 4 m: perimeter = **22** m

Part 8 Word problems What does the remainder mean in the story?

79) 26 kids go on a trip. Each van holds 6 kids. How many vans do they need?

Number sentence **26 ÷ 6 = 4 R 2, so 5 vans**

Answer **5** vans

Draw or show your work

80) Mia shares 30 stickers equally with 4 friends. How many stickers does each friend get?

Number sentence **30 ÷ 4 = 7 R 2, 7 each**

Answer **7** stickers

Draw or show your work

81) Ben puts 45 eggs in cartons of 12. How many eggs are left over?

Number sentence **45 ÷ 12 = 3 R 9**

Answer **9** eggs

Draw or show your work

Division with Remainders: Practice (continued)

Part 8 Word problems (continued)

82) A ribbon is 50 inches long. Kim cuts pieces 8 inches long. How many whole pieces can she cut?

Number sentence **50 ÷ 8 = 6 R 2, 6 pieces**

Answer **6** pieces

Draw or show your work

83) Challenge: 58 books go on shelves that hold 9 each. How many shelves are full, and how many books are left?

Number sentence **58 ÷ 9 = 6 R 4**

Answer **6 R 4**

Draw or show your work

Check-In: Division with Remainders

Show what you know. No help this time.

Mastery: 8 of 9 (80%)

Part 1 Make groups

1) **Circle groups of 3.**



6 groups of 3, 1 left over

Part 2 Use a times fact

2) **14 ÷ 3: the biggest fact up to 14**

3 × 4 = 12

14 − 12 = 2

so 14 ÷ 3 = 4 R 2

3) **17 ÷ 4: the biggest fact up to 17**

4 × 4 = 16

17 − 16 = 1

so 17 ÷ 4 = 4 R 1

Part 3 Divide

4) 18 ÷ 4 = 4 R 2

5) 21 ÷ 4 = 5 R 1

6) 77 ÷ 9 = 8 R 5

7) 73 ÷ 9 = 8 R 1

8) 11 ÷ 3 = 3 R 2

9) 51 ÷ 9 = 5 R 6